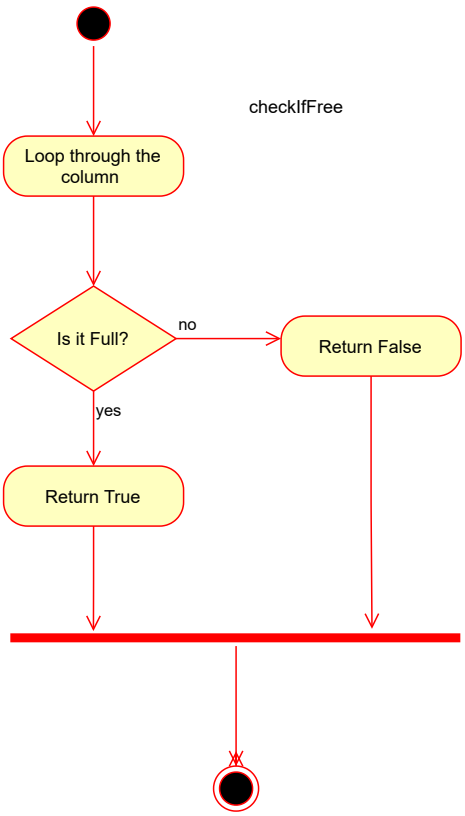
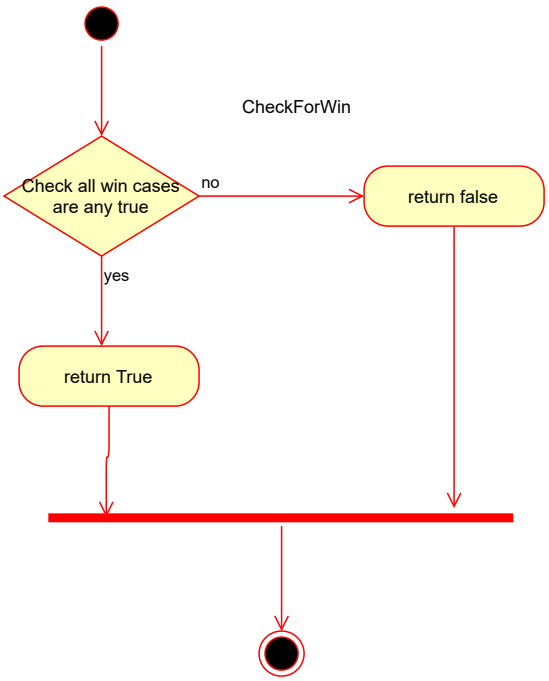
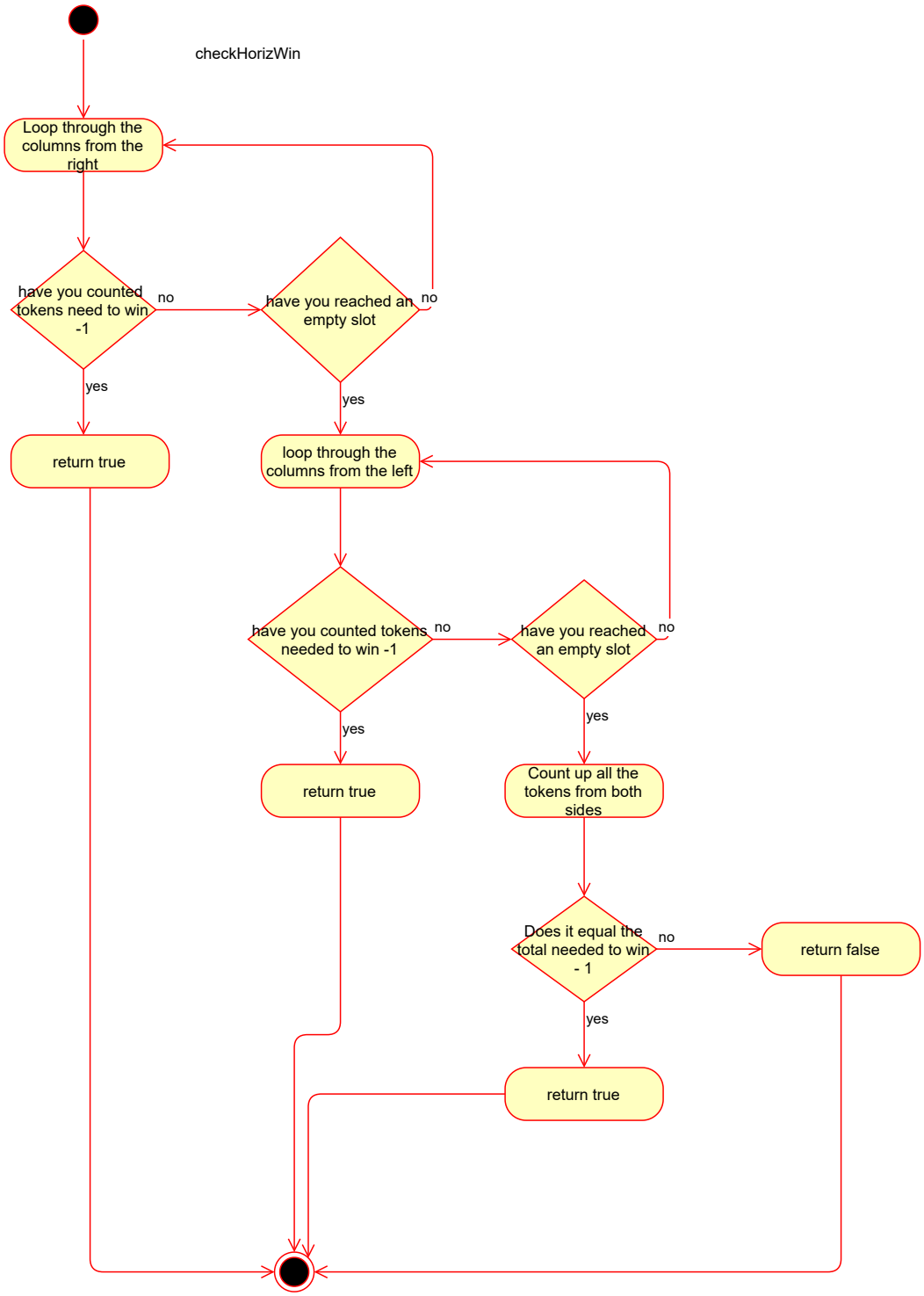
IGameboard

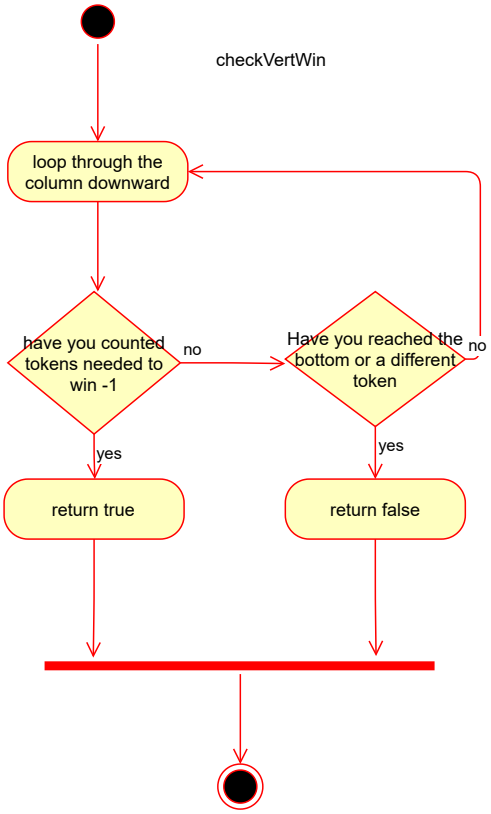

checkIfFree
Loop through the column
Is it Full?
no
Return False
yes
Return True

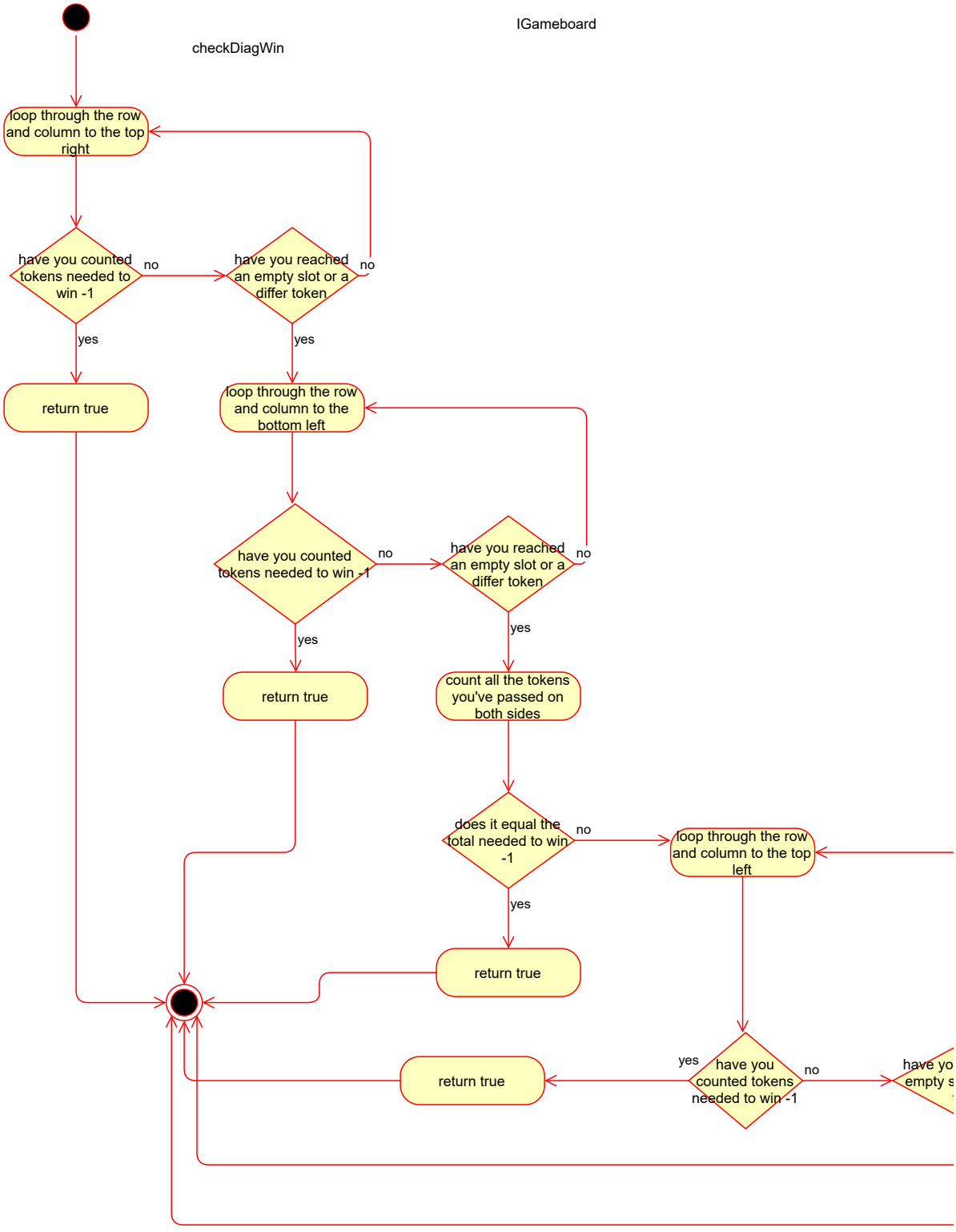

CheckForWin
Check all win cases are any true
no
return false
yes
return True

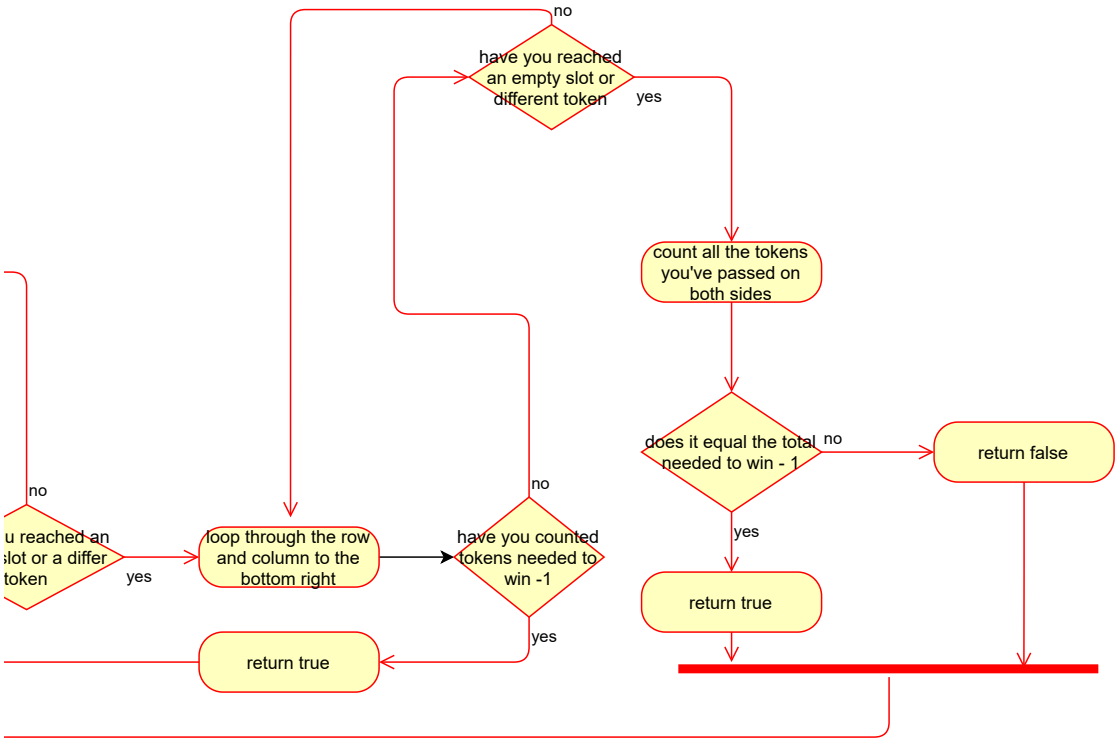
IGameboard

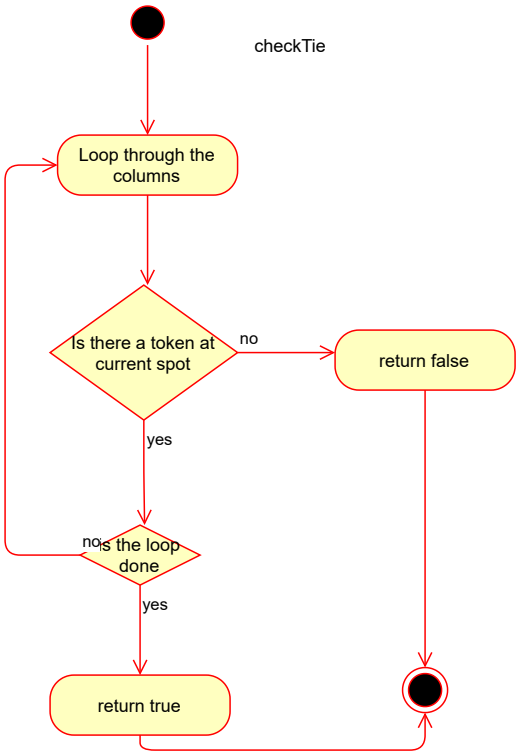

checkHorizWin
Loop through the columns from the right
have you counted tokens need to win -1
no
have you reached an empty slot
no
yes
yes
return true
loop through the columns from the left
have you counted tokens needed to win -1
no
have you reached an empty slot
no
yes
yes
return true
Count up all the tokens from both sides
Does it equal the total needed to win - 1
no
return false
yes
return true

IGameboard


checkVertWin
loop through the column downward
have you counted tokens needed to win -1
no
Have you reached the bottom or a different token
no
yes
yes
return true
return false


IGameboard
checkDiagWin
loop through the row and column to the top right
have you counted tokens needed to win -1
no
have you reached an empty slot or a differ token
no
yes
yes
return true
loop through the row and column to the bottom left
have you counted tokens needed to win -1
no
have you reached an empty slot or a differ token
no
yes
yes
return true
count all the tokens you've passed on both sides
does it equal the total needed to win -1
no
loop through the row and column to the top left
yes
return true
yes
have you counted tokens needed to win -1
no
return true


no
have you reached an empty slot or different token
yes
count all the tokens you've passed on both sides
does it equal the total needed to win - 1
no
return false
yes
return true
u reached an lot or a differ token
no
yes
loop through the row and column to the bottom right
have you counted tokens needed to win -1
no
yes
return true


checkTie
Loop through the columns
Is there a token at current spot
no
return false
yes
no
is the loop done
yes
return true